

Midterm CPIT-285 (first semester 2024)

From ch1 to ch6

- 1- In OpenGL a 2D region on a 3D object is called
 - a- Face
 - b- Vertex
 - c- Edge
 - d- None of above
- 2- Raster graphics?
 - a- Lines of pixels
 - b- Mathematical formula
 - c- None of above
- 3- Virtual reality?
 - a- Physical object unreal environment

3. Which of the following is used to reset the graphic window size?

- a. `glutReshapeFunc (reshape)`
- ☒ b. `glutReshapeWindow (reshape)`
- c. `glutPositionFunc (reshape)`
- d. `glutPositionWindow (reshape)`

4. In the midpoint algorithm, if the midpoint is below the line, then the next pixel is:

- a. (x_k+1, y_k+1)
- ☒ b. (x_k+1, y_k)
- c. (x_k, y_k)
- d. (x_k, y_k+1)

$f < 0 \rightarrow =$

Graphics card parts:

1-

2-

3-

4-

Fill the plank : Electron beam in the cathode are negative ,the anode is positive

CRT techniques: :

1-

2-

3-

Write the line drawing analytical method?

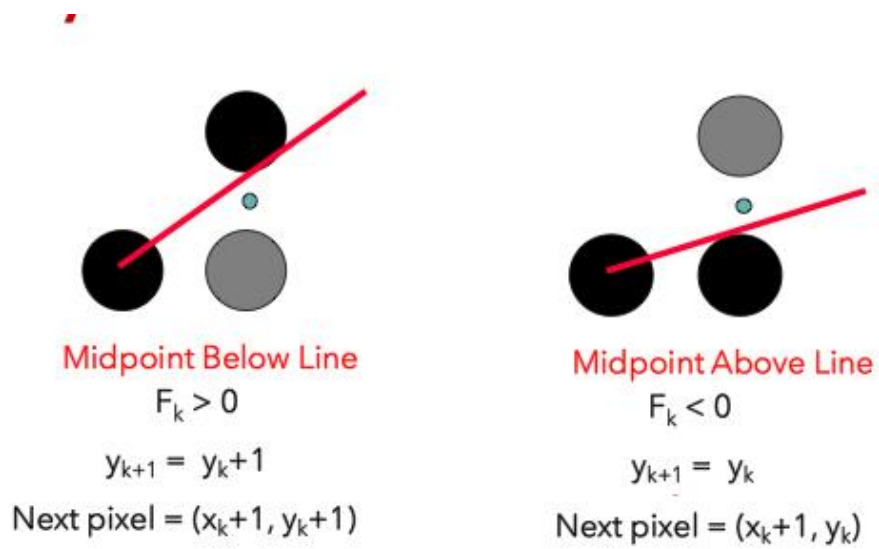
```
//Given (ax, ay) and (bx, by)

double m = (double) (by-ay) / (bx-ax) ;
double c = ay - m*ax;
double y;
int    iy;
for (int x=ax; x<=bx; x++) {
    y = m*x + c;
    iy = round(y);
    setPixel(x, iy);
}
```

Rasterization problem:

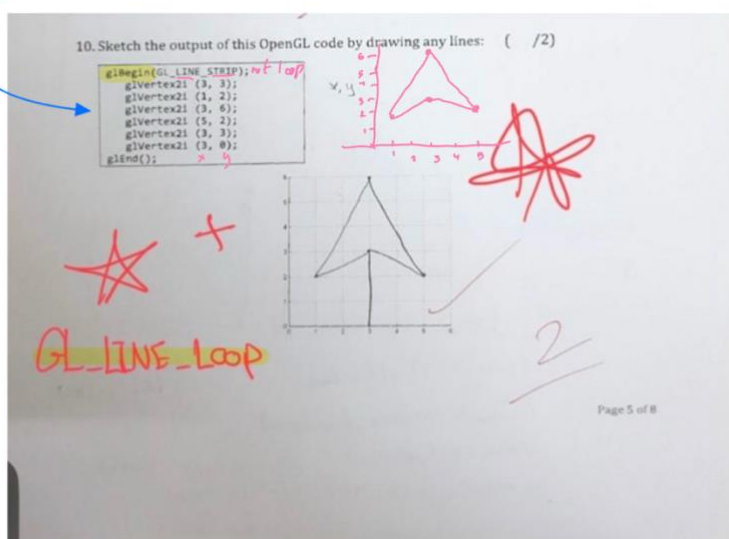
Given only the two end points how to compute the intermediate pixels so that the set of pixels closely approximate the ideal line

Draw line mid point algorithm?

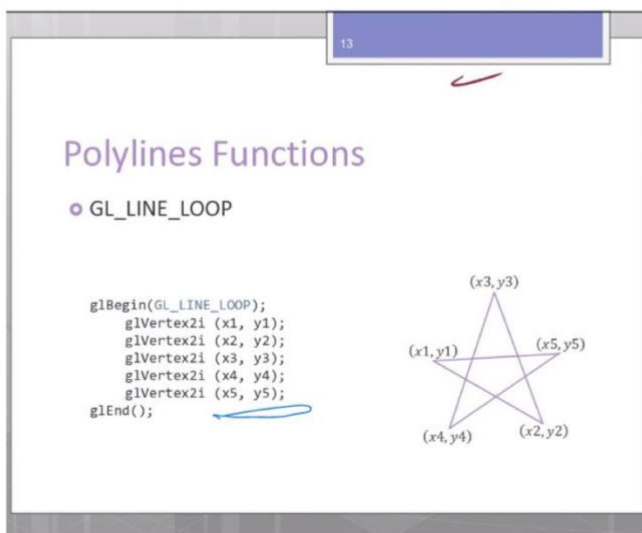


The difference between frame rate and refresh rate and their units ?

السؤال الثاني: جابت احداثيات وانت ترسمين رسمة، الذي قطع مثل للاديس



الثانية قطع زجاجة، كلها تقعد على اللبؤد الموجودة في الكود، راجعوا زين!



Write the following codes:

a- Create 200 pixels and 150 pixels window, with Coordinate(75,20)

b- Draw red triangle in the middle of the window

c- Set the window background color to gray

What are the five coordinate stages and define one of your choice?

Fill the plank

- 1- The middle layer contains in LED
- 2- The middle layer contains in LCD

Good luck<3

لا تنسونا من دعواتكم